

DEBOJYOTI CHATTORAJ

ADDRESS:

Annaiah Reddy Road, Kamadhenu Layout, Konnappa Agrahara, Electronic City, Bangalore - 560100

Email: debojyotichattoraj1996@gmail.com

Mobile: +(91) 8900731996

Portfolio: [Portfolio](#) | **LinkedIn:** [LinkedIn](#)

CAREER OBJECTIVE:

Aspiring Data Scientist, Data Analyst and Machine Learning Engineer leveraging 2+ years of hands-on experience in model development, deployment, and full-stack integration. Eager to apply deep learning, data engineering, and cloud-native MLOps practices to drive actionable insights in pharmaceutical and life-science domains.

PROFESSIONAL SUMMARY:

Multidisciplinary AI/ML professional skilled in designing, training, and optimizing supervised and unsupervised models using TensorFlow, Keras, Scikit-Learn, and OpenCV. Proficient in data preprocessing, feature engineering, and hyperparameter tuning to achieve 85%+ validation accuracy on complex classification tasks. Experienced in containerized deployments with Docker, Flask-based REST APIs, and Google Cloud Platform. Strong background in pharmaceutical R&D ensures data integrity and compliance throughout the ML lifecycle.

CORE COMPETENCIES:

Python | SQL | Supervised & Unsupervised Learning | Deep Learning (CNN, RNN) | Transfer Learning | Computer Vision & OpenCV | Data Wrangling (Pandas, NumPy) | Model Deployment & Monitoring | Hyperparameter Tuning | Flask API Development | Docker & Kubernetes Basics | Frontend Development (HTML, CSS & Bootstrap) | Data Visualization (Matplotlib, PowerBI) | Google Cloud Platform (Compute Engine, Storage) | PostgreSQL | Anaconda | Jupyter Lab | Data Analysis (PowerBI) | Excel (Lookup, VBA, Macros) | ISO Standards Implementation | Computerized System Validation | SDLC | Scrum | Agile, Waterfall & V methodology | CI/CD for ML (Git, GitHub Actions) | Six Sigma Green Belt | Version Control (Git)

- Comprehensive understanding of supervised, unsupervised, and reinforcement learning algorithms, including regression, decision trees, clustering, and deep neural networks.
- Knowledge of model evaluation metrics and validation techniques such as cross-validation, confusion matrix analysis, ROC-AUC, precision, recall, and F1-score.
- Core proficiency in data preprocessing and feature engineering, including handling missing values, encoding categorical variables, normalization, and dimensionality reduction (PCA).
- Skilled in Hyperparameter tuning strategies (grid search, random search, Bayesian optimization) to enhance model performance and prevent overfitting.
- Proficient in designing and training deep learning architectures (CNNs, RNNs, Transformers) using TensorFlow, Keras.
- Experienced in computer vision tasks—image classification, object detection, and segmentation—leveraging OpenCV and convolutional neural networks.
- Adept at deploying machine learning models as RESTful services with Flask or FastAPI and containerizing applications using Docker for scalable production environments.
- Implemented MLOps pipelines with CI/CD workflows via GitHub Actions, experiment tracking using MLflow, and model/version control with DVC.
- Familiar with cloud-based ML workflows on GCP (AI Platform), enabling scalable training, serving, and monitoring.

ORGANIZATIONAL EXPERIENCES:

1. **Thermo Fisher Scientific India Pvt. Ltd, Bangalore**
Designation: Technical Application Specialist (From 6th July 2023 to Present)
2. **Spinco Biotech Pvt. Ltd. (Shimadzu Asia Pacific)**
Designation: Executive Application Support (From 1st Sep 2022 to 1st July 2023)
3. **Wipro Limited, Bangalore**
Designation: Scientist (From 1st Jul 2021 to 30th Aug 2022)
4. **Edward Food Research and Analysis Centre Pvt. Ltd., Kolkata**
Designation: Jr. Chemist (From 1st October 2019 to 30th June 2021)
5. **Alkem Laboratories Pvt. Ltd., Sikkim**
Designation: QC Officer (From 2nd August 2018 to 15th September 2019)

SELECTED PROJECTS:

1. Automated Machine Learning (AutoML):

Developed and deployed an AutoML web application on Google Cloud Run that enables users to upload CSV datasets, select target variables, and automatically train classification or regression models with built-in preprocessing, feature engineering, and hyperparameter tuning. Integrated Google Cloud Storage for dataset/model management and implemented automated evaluation with HTML/PDF reporting.
For more details, please visit 'My Portfolio' section of <https://portfolioofdebojyoti.in>.

2. Flower Image Classification

The classification of flowers based on their images is a practical and visually appealing application of deep learning in the domain of computer vision. This project presents the development of a flower identification system using a pre-trained ResNet50 convolutional neural network model, fine-tuned on a flower dataset.
For more details, please visit 'My Portfolio' section of <https://portfolioofdebojyoti.in>.

3. Blog Website

Developed a full-stack blogging platform enabling users to create, edit, and comment on posts, with role-based access controls and input validation.

- Crafted a responsive, mobile-first UI using Bootstrap, ensuring consistent typography, grid layouts, and component styling across devices.
- Engineered a RESTful backend with Python Flask, structuring modular Blueprints for authentication, content management, and API endpoints.
- Deployed the application to an Oracle Cloud VPS (Ubuntu), leveraging Caddy as a secure reverse proxy (HTTPS/TLS) and Gunicorn as the WSGI server for scalable request handling.
- For more details, please visit 'My Portfolio' section of <https://portfolioofdebojyoti.in>.

EDUCATIONAL QUALIFICATIONS:

- **Bachelor of Pharmacy (B. Pharm)** – Maulana Abul Kalam Azad University of Technology (2014-18) | YGPA: 7.94
- **Higher Secondary (Class 12)** – Saidabad Manindra Chandra Vidhyapith | WBCHSE (2013) | 69.20%
- **Secondary (Class 10)** – Saidabad Manindra Chandra Vidhyapith | WBBSE (2011) | 82.50%

TECHNICAL SKILLS:

1. Defect Management Tools (JIRA & Azure DevOps)
2. Languages & Frameworks: Python, SQL, TensorFlow, Keras, Scikit-Learn, OpenCV, Pandas, NumPy, Matplotlib
3. Web & API: Flask, HTML5, CSS3, Bootstrap
4. Databases & Storage: PostgreSQL, MySQL, SQLite
5. DevOps & Cloud: Docker, Google Cloud Platform
6. Virtualization Tools (Hyper-V & Azure DevTestLab)
7. Operating System (Linux, Windows, Windows server 2019/2022)
8. Tools & IDEs: Jupyter Lab, Anaconda, VS Code
9. Structured Query Language (SQL)
10. Visualization & Reporting: PowerBI, MS Office

CERTIFICATIONS:

- 100 Days of Code – The Complete Python Pro Bootcamp from Udemy
- SQL Essential Training from NASBA
- Python Basics from HackerRank
- SQL Basics from HackerRank
- Certification course on Lean Six Sigma Green Belt Certification (IASSC) from Simplilearn.
- Certification course on Agile Scrum Foundation from Simplilearn.
- Certification course on Agile Methodology from GreatLearning.

PROFESSIONAL STRENGTH:

Team Player | Time Management | Motivator | Go-getter